PhD Project Proposal - Using genomics to evolve our understanding of the phylogeographic history of the Galapagos giant tortoise radiation

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Project Overview

Background

The study of diversification of living entities, ultimately leading to the definition of species and understanding of the relationships between them, is a key part of evolutionary biology and systematics studies. This process of speciation is a product of several processes such as gene flow, mutations, selection, adaptation, and drift. As populations are almost continuously subjected to these processes, the process of speciation is dynamic as well.

Understanding the limits of species is crucial to understanding how to conserve them. The field of taxonomy draws on different traits — morphological, behavioural, ecological, and more recently, genetic (Padial *et al.*, 2010). Over time, a diversity of approaches for defining species have been proposed, including on reproductive compatibility (Wright, 1940; Paterson, 1985; Mayr, 1999), isolation (Dobzhansky, 1970; Mayr, 1999), evolutionary tendencies (Simpson, 1951; Wiley, 1978), shared characteristics (Mishler and Brandon, 1987), or genotypic clusters (Mallet, 1995). The "Unified Species Concept" (De Queiroz, 2007) explains that the various species definitions are variants of the overall concept of species as 'separately evolving metapopulation lineages'. Therefore, demonstrating evidence of lineage divergence and separation would indicate the existence of separate species.

The involvement of genetic data has helped investigate lineage separation and identify differences that might not otherwise be found through morphological or behavioural studies alone, e.g., cryptic species (Adams *et al.*, 2014, 2014; Shin and Allmon, 2023). Data from genomes also facilitates the study of relationships between species, from building trees to depict evolutionary relationships between species to estimating the times and rates of their divergence (Young and Gillung, 2020). However, even molecular methods have their complexities. It is important that studies consider balanced and adequate sampling of individuals as well as loci to inform their analyses. Additionally, bioinformatic software used in genomic studies often require the user to define thresholds parameters which play critical roles in the analyses and can pose the risk of error in the case of incorrect definitions (Spelman and van Arendonk, 1997; Puechmaille, 2016; Dattani *et al.*, 2022).

Genomic approaches for the study of biodiversity can involve a wide range of methods from short sequences acting as barcodes for species identification, to DNA from organelles such as mitochondria, reduced representation sequencing, and whole genome sequencing (Theissinger *et al.*, 2023). Occasionally, different types of markers can yield differing signals. One such example is the occurrence of mitonuclear discordance, or a mismatch between patterns of differentiation in mitochondrial and nuclear DNA. This is due to the difference in mechanisms of evolution and inheritance in mitochondrial and nuclear DNA. Mitochondrial genomes are haploid, usually maternally inherited, and do not undergo recombination during inheritance (Zink and Barrowclough, 2008). Mitochondrial genes also reflect a fourfold smaller effective population size than nuclear DNA, and so fixation of new mutations as well as lineage sorting occurs much faster for mitochondrial genes (Funk and Omland, 2003; Tautz *et al.*, 2003; Zink and Barrowclough, 2008; Toews and Brelsford, 2012). This can lead to homoplasy or mitochondrial capture in cases of hybridisation between species. These instances are especially seen in lineages with complex evolutionary histories involving processes (often a combination of multiple) such as alternations

between isolation and gene flow (e.g., secondary contact), invasions, local adaptation, incomplete sorting of ancestral polymorphisms, and sex-biased dispersal (Funk and Omland, 2003; Zink and Barrowclough, 2008; Toews and Brelsford, 2012). Often, this discordance also has biogeographic patterns.

Study System

Islands provide a dynamic system to study evolutionary processes. Evolution in islands is expected to be through a cascade of dispersal events, vicariance, or both (Poulakakis *et al.*, 2012), and populations from different islands typically evolve independently. This leads to strong biogeographic signals and a complex evolutionary history. After

The Galapagos Islands are a notable example of such a system, and renowned in the study of evolution of species. The islands, located 900 km off the western coast of South America, were formed due to the south-eastwards movement of the Nazca plate over basaltic plumes (Naumann and Geist, 2000; Geist *et al.*, 2014), and so the island ages decrease towards the west. There has also been submersion, splitting, and merging of islands (Ali and Aitchison, 2014). This, along with changing lava flows, add further complexity to the geological history the islands and therefore also the evolutionary histories of its species.

The Galapagos Giant Tortoises (*Chelonoidis* spp. Fitzinger, 1835) encompass 16 taxa (three of which are extinct), with at least 7 – 9 extant species (Gaughran *et al.*, 2023). Based on patterns in the mitochondrial genome, the diversification of these taxa has been found to closely follow the history of the island system (Poulakakis *et al.*, 2012). While there have been several studies investigating the history of the Galapagos Giant Tortoises (Caccone *et al.*, 1999, 2002; Ciofi *et al.*, 2002; Beheregaray *et al.*, 2003, 2004; Russello *et al.*, 2005; Ciofi *et al.*, 2006; Poulakakis *et al.*, 2012; Edwards *et al.*, 2014; Poulakakis *et al.*, 2015; Garrick *et al.*, 2015; Miller *et al.*, 2018; Poulakakis *et al.*, 2020; Jensen, Gaughran, *et al.*, 2022; Gaughran *et al.*, 2023), species delimitation has been a challenge. This is largely due to the recent divergence times between the lineages and complex biogeography.

Much of the research into the Galapagos Giant Tortoise system had utilized mitochondrial and microsatellite data. The recent advent of next generation sequencing and the use of genome-wide polymorphic markers revealed new insights into the diversification history of the lineages that had not been seen through mitochondrial and microsatellite data (Miller *et al.*, 2018; Jensen, Gaughran, *et al.*, 2022; Jensen, Quinzin, *et al.*, 2022). Differences in microsatellite-based and nuclear marker-based analyses have been seen in other studies as well where microsatellite markers have been seen to have a risk of homoplasy with their high mutation rates (Morin *et al.*, 2004). Nuclear markers such as single nucleotide polymorphisms (SNPs), with lower mutation rates are more suitable for studies of both strongly and weakly diverged populations (Jeffries *et al.*, 2016), and provide a greater number of informative loci.

So, in my study, I aim to utilize whole genome, especially nuclear, data to investigate the gaps in the understanding of the taxa in the Giant Tortoise system. The overall goal for my project is to understand the mechanisms at play affecting species in island systems. Through my chapters, I will scale up from population level studies in an individual island to a larger picture of the entire system through nuclear data.

Data Chapter 1: Conservation genomics of Southern Isabela Island Tortoises

While for most islands, one species is associated with a single island, Isabela and Santa Cruz islands host multiple distinct lineages.

Southern Isabela hosts two species *C. guntheri* (Baur 1889) and *C. vicina* (Günther 1874) across several population fragments (Edwards *et al.*, 2014). Mitochondrial DNA reflects low levels of

genotypic structuring between the populations distributed along Southern Isabela (Ciofi *et al.*, 2006) while microsatellite data reveals three genetic clusters which correspond to geographic localisations (Edwards *et al.*, 2014). These results correspond with a gradient in morphological characteristics from West to East (Fritts, 1984). *Chelonoidis guntheri* and *C. vicina* are the most recently diverged amongst the system (Poulakakis *et al.*, 2012) formed through radiation within Isabela and isolation due to formation of lava bridges (Ciofi *et al.*, 2006). There are also signs of recent gene flow and migrations between populations (Ciofi *et al.*, 2002, 2006; Beheregaray *et al.*, 2004; Edwards *et al.*, 2014). These factors complicate the disentanglement of the evolutionary history and taxonomy of these populations and bring forth the question of whether the two species are indeed different taxonomically or whether the populations represent a gradient in one species.

Additionally, for the conservation of the species endemic to Southern Isabela, it is critical to understand the genetic health of their constituent populations. This is especially of importance in the case of *C. guntheri* which is critically endangered (IUCN, 2015) due to exploitation as a food source and poaching (Cayot and Lewis, 1994; Powell and Gibbs, 1995; Pritchard, 1996), and it would be critical to identify how many populations remain and assess their health from a genetic viewpoint.

Recent, intensive survey efforts in Southern Isabela by the Galapagos National Park and Galapagos Conservancy have resulted in a comprehensive set of >400 blood samples from tortoises across the 12 population fragments in Southern Isabela. These samples were imported to Newcastle University in 2024 and are available for my research.

The aims of this chapter are to:

1. Investigate the unique genetic clusters (if any) of *C. guntheri* and *C. vicina* in the Southern region of Isabela Island
2. Understand the demographic histories of the populations and the instances of divergence between them
3. From a genetic standpoint, determine which populations are in need of urgent conservation attention and action

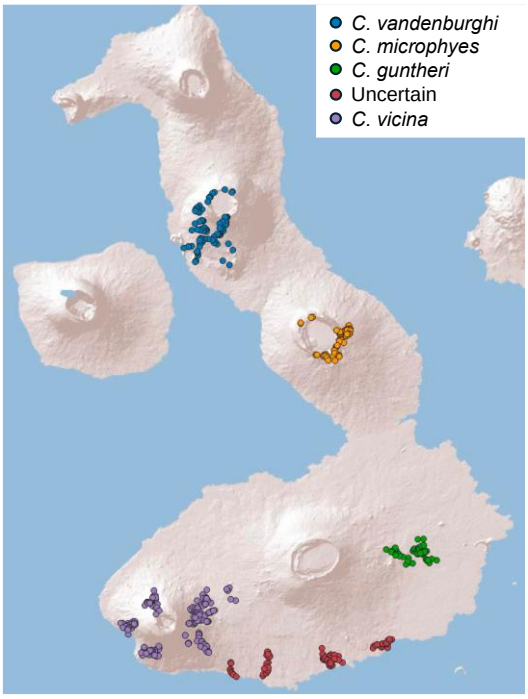


Figure 1. Map of Isabela Island showing the sampling locations to be used in this study. In red are the population fragments with uncertain taxonomic affiliation

To understand levels of within and between population diversity among the population fragments, I will sequence full genomes from five individuals from each population fragment to a depth of ~10X. This data will be combined with an existing dataset of 40 genomes already collected from previous studies. The sequences obtained will be trimmed, aligned (to the *C. abingdonii* reference genome (Quesada *et al.*, 2019), and filtered using the BAM pipeline in PaleoMIX (Schubert *et al.*, 2014). Genotype calling will be done through the mpileup/call pipeline in BCFtools (Danecek *et al.*, 2021), and the resulting genotypes filtered using VCFtools (Danecek *et al.*, 2011).

This would be followed by clustering analyses (e.g., ADMIXTURE (Alexander, Novembre and Lange, 2009), PCA) to identify distinct population clusters, and tree-building algorithms for phylogeny (such as maximum likelihood trees in IQ-Tree (Minh *et al.*, 2020) and Bayesian inference trees in BEAST2 (Bouckaert *et al.*, 2019)) to understand the relationships between them.

Subsequently, investigation of divergence times of populations (also possible in BEAST2) and analyses of current day gene flow (BAYESASS (Wilson and Rannala, 2003)) will help define the demographic histories of the populations. The events of divergence and gene flow can also be modelled through TreeMix (Pickrell and Pritchard, 2012). Additionally, GPS locations of individuals were noted during the survey and can be used to understand geographical ranges of individual fragments, contributing to the understanding of potential routes for gene flow between fragments.

For the third objective, I aim to estimate recent effective population sizes of the distinct populations (through coalescence based methods or Approximate Bayesian Computation), as well as their levels of within-population genetic diversity (through statistics such as π (Nei and Li, 1979)). The recent effective size can give insight towards the impact of genetic drift on the population, the rate of loss of heterozygosity, selection acting on a new mutation, as well as the mean time to extinction (Kimura and Crow, 1963; Lande, 1994; Charlesworth, 2009). This will help identify any populations with consistently low effective population sizes or low diversity, hence identifying them to be in need for urgent conservation action.

Data Chapter 2: Investigating Mitonuclear Discordance in the Relationships Between the Two Santa Cruz Island Tortoise Species

A similar lineage complexity like in Southern Isabela lies in the island of Santa Cruz where conversely, what was thought to be two populations of one species was resolved into two species (*C. porteri* (Rothschild, 1903), *C. donfaustoi* (Poulakakis, Edwards & Caccone, 2015)) after studies using microsatellite and mitochondrial data (Beheregaray *et al.*, 2003; Poulakakis *et al.*, 2015). However, recent studies (Jensen, Gaughran, *et al.*, 2022; Gaughran *et al.*, 2023) have shown that, while the mitochondrial phylogenies of the species show a deep divergence, the nuclear phylogenies place them in the same clade. This is an example of mitonuclear discordance which is also seen in other lineages amongst the Galapagos Giant Tortoise system (Jensen, Gaughran, *et al.*, 2022).

Mitonuclear discordance can be a result of numerous processes, such as introgression, incomplete lineage sorting, sex-biased dispersal, etc (Zink and Barrowclough, 2008; Toews and Brelsford, 2012). *Chelonoidis donfaustoi* is listed as critically endangered in the IUCN Red List (IUCN, 2016), hence making it imperative to resolve the evolutionary history of the two populations on Santa Cruz to resolve their identities as individual species or to synonymise them. Furthermore, it would be important to clarify the extent of impact by human settlements — to understand if the low levels of hybridisation can be attributed to human impact or if the pattern goes further back in time.

The aims for this chapter are to:

1. Resolve the evolutionary histories of the two giant tortoise taxa on Santa Cruz
2. Reinvestigate the species statuses of *C. porteri* and *C. donfaustoi* based on levels of genetic differentiation

For this chapter, I will use a combination of whole genome sequencing and reduced representation genome sequencing (e.g., double digest restriction-site associated DNA or DDRadSeq) of samples obtained during previous surveys in order to increase the sample size for the analyses. I aim to have both nuclear and mitochondrial data for each individual.

The initial steps will be determining the genetic structuring present across the populations, ancestries of the individual samples, and an overview of the relationships between the genetic clusters through cluster-based analyses such as STRUCTURE (Pritchard, Stephens and Donnelly, 2000), ADMIXTURE (Alexander, Novembre and Lange, 2009), principle component analysis (PCA), discriminant analysis of principle components (DAPC) (Jombart, Devillard and Balloux, 2010). This

can identify genetic clusters that could have formed through distinct evolutionary histories or through isolation by distance. Following this, species delimitation will be assessed through multiple approaches such as the multispecies coalescent models of SNAPP in the BEAST2 software (Bouckaert *et al.*, 2019) and Bayesian Phylogenetics and Phylogeography (Flouri *et al.*, 2018), heuristic methods such as *gdi* (Jackson *et al.*, 2017), or delineate (Sukumaran, Holder and Knowles, 2021). Phylogenetic analyses will be carried out through Bayesian (e.g., MrBayes (Ronquist *et al.*, 2012)) or Maximum-Likelihood (e.g., RAxML (Stamatakis, 2014); IQ-TREE (Minh *et al.*, 2020)) approaches.

To understand the mitonuclear discordance, I have designed my proposed methodology around a combination of approaches used in similar studies conducted in different species complexes (Singhal and Moritz, 2012; Cahill *et al.*, 2013; Phuong, Bi and Moritz, 2017; Ivanov, Lee and Mutanen, 2018; Firneno *et al.*, 2020; Marshall *et al.*, 2021; Thomas J Firneno *et al.*, 2021; DeRaad *et al.*, 2023).

As a starting point, I will look into the levels of sequence divergence between the lineages in both mitochondrial and nuclear DNA to identify the differential divergence patterns between the two. This would include summary statistics such as F_{ST}, Tajima's D, D_{XY}, and π (Wright, 1943; Nei and Li, 1979; Tajima, 1989) which can be done through software such as pixy (Korunes and Samuk, 2021). As the Galapagos Giant Tortoise diversification is recent, the discordance observed could be due to introgression, gene flow with differential selection, or incomplete lineage sorting. I will carry out tests for selection (e.g., Sweepfinder2 (DeGiorgio *et al.*, 2016)) to determine whether the divergence could be linked to local adaptation, hence maintaining the divergence in the face of gene flow. To understand the effect of introgression (if any), four-taxon D-statistic test (Green *et al.*, 2010) and locus-specific introgression tests can be used. In the case of lack of signs of recent gene flow, it is likely that the discordance is due to incomplete lineage sorting. Lastly, to look into the histories of the populations, I aim to estimate the demographic history through Approximate Bayesian Comparison methods, TreeMix (Pickrell and Pritchard, 2012), species distribution models, rangeExpansion (Peter and Slatkin, 2013), and estimated effective migration surfaces (EEMS) (Petkova, Novembre and Stephens, 2016) to understand which processes could have driven the establishment of the mitonuclear discordance, and possibly understand when they began.

Data Chapter 3: A Comprehensive Biogeographic Model for the Radiation of Galapagos Giant Tortoises

My third chapter will investigate and establish a well-supported account of the biogeography of the Galapagos Giant Tortoises using nuclear genomes.

A previous study (Poulakakis *et al.*, 2020) used complete or near-complete mitogenomes to investigate the colonisation history and phylogeny of the Giant Tortoise taxa across the archipelago. With the evidence of instances of mitonuclear discordance in the Galapagos Giant Tortoises, it would be pertinent to study the phylogeny and radiation using nuclear data, ultimately developing a model that is in accordance with the patterns obtained from both. A recent study with nuclear data (Gaughran *et al.*, 2023) aimed to evaluate the species delimitations of the Galapagos Giant Tortoises under the multispecies coalescent framework, and found evidence for at least 7 – 9 extant species. This acts as a starting point for the phylogenetic reconstruction I aim to do.

The aims for this chapter are to:

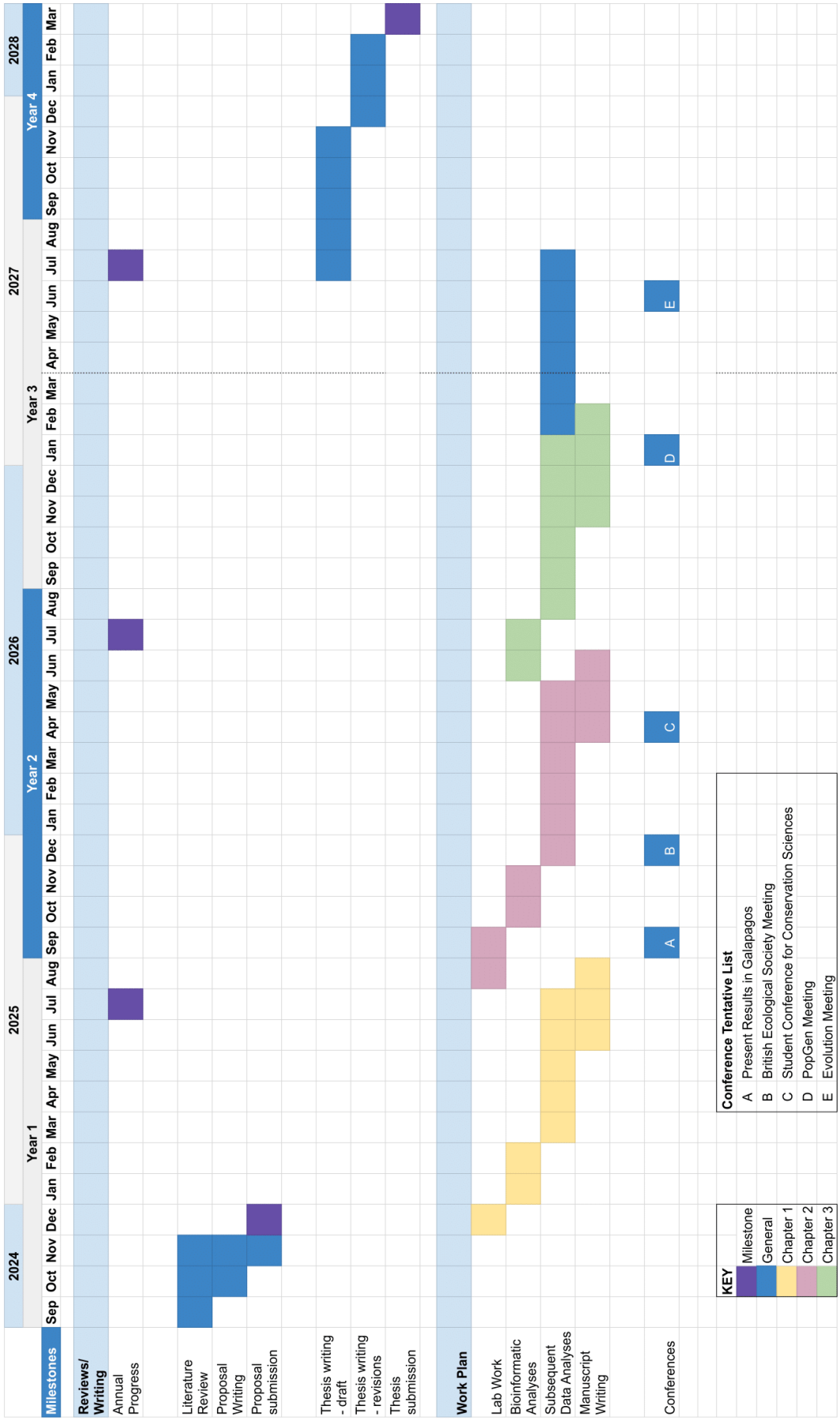
1. Reconstruct a time-calibrated phylogenetic tree for all Galapagos Giant Tortoise lineages
2. Determine the ancestral ranges of the lineages
3. Assimilate models derived from nuclear and mitochondrial data into a comprehensive model, aligned with the geological history of the Galapagos Archipelago

For this chapter I will use the existing dataset of ~13 genomes per extant lineage, and exemplar sequences from the extinct lineages. Identifying the individual lineages and the relationships through them can be possible through construction of dated phylogenies (possible through BEAST2 (Bouckaert *et al.*, 2019)). The ancestral ranges of each lineage will be reconstructed using the program BioGeoBEARS (Matzke, 2013) through which a range of biogeographic models can be tested. Akaike Information Criterion (AIC) can then be used to determine the best-fitting model. This nuclear biogeography, compared to established mtDNA-based trees can help develop a more complete picture of the evolutionary history of the Galapagos Tortoises. To further enhance the picture, it would be of interest to align the evolutionary models obtained with the geological models of the archipelago (e.g., (Geist *et al.*, 2014) in order to track the co-occurrence of salient events with the changes in the island landmasses such as island emergences and submergences, as well as lava flows and flooding which can connect or separate populations.

Research Implications

By the end of the project, I hope to have added to the knowledge of Galapagos Giant Tortoise evolution. My results will help add clarity to the history of the taxa in southern Isabela, in Santa Cruz, and also add to what is known about the biogeography of the entire system. The mechanisms uncovered through this study could also provide insight into potentially similar processes in other species in the Galapagos, as well as other island systems. Furthermore, with several populations in various species becoming fragmented over time due to habitat and population loss, radiation, isolation, and vicariance processes in islands can inform future directions in very fragmented population systems. A significant outcome of this study would be understanding what processes might lead the mitonuclear discordance in the Giant Tortoises of Galapagos, which would also be part of the greater knowledge base of mitonuclear discordance in other species too. Lastly, this project demonstrates the significance of genomic studies in understand evolution and speciation, as well as its implications for the conservation of species.

Gantt Chart


2024
2025
2026
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Year 1
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Year 3
Year 4
Sep Oct Nov Dec Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec Jan Feb Mar
Milestones
Reviews/ Writing
Annual Progress
Literature Review
Proposal Writing
Proposal submission
Thesis writing - draft
Thesis writing - revisions
Thesis submission
Work Plan
Lab Work
Bioinformatic Analyses
Subsequent Data Analyses
Manuscript Writing
Conferences
A
B
C
D
E
KEY
Milestone
General
Chapter 1
Chapter 2
Chapter 3
Conference Tentative List
A Present Results in Galapagos
B British Ecological Society Meeting
C Student Conference for Conservation Sciences
D PopGen Meeting
E Evolution Meeting

Ethical Approval

In progress

Budget

Year 1
£500 for registration at a genome analysis and bioinformatics training workshop.
£3000 for medium coverage sequencing of Southern Isabella individuals (n=20, £150 per sample)

Year 2
£500 for travel and registration at a UK national conference (e.g., British Ecological Society, PopGroup) with attendance at satellite workshops on science communication and/or analysis methods.
£3000 for ddRAD of Santa Cruz samples
£2000 for travel to the Galapagos to report results from Chapter 1 to Galapagos organizations

Year 3
£1000 for travel and registration at an international conference (e.g., Evolution, European Society of Evolutionary Biology) with attendance at satellite workshops on science communication and/or analysis methods.

Training Needs Analysis

Objective	How will this be measured?	Achieved?
Gain basic knowledge and proficiency in Python Programming Language	I will have achieved this upon accomplishment of one or more workshops or training programs for using Python	No
If not achieved: The workshops I wish to attend are coming up so I am yet to attend them.		
Gain experience in understanding phylogenomic analyses, with a focus on building gene and species trees.	I will attend workshops to undergo training specific to phylogenomics, and apply the knowledge gained to my own research	No
If not achieved: The workshops I wish to attend are coming up so I am yet to attend them.		
Learn how to do science communication and outreach as a means to talk about research in my field and.	I will gain experience in events which allow for science communication and research.	No

Risk Analysis

Risk	Mitigation
Insufficient data or poor quality from sequencing results	Ensure presence of enough sample quantity to allow for carrying out sequencing again, e.g., samples collected from Southern Isabela are present in Newcastle University with sufficient quantity.
Computer failure	Keep a backup of important files in a hard drive. Ensure to get computer repaired without delay while using personal PC in the meantime.
Server failure or similar issues with the high-performance computing cluster	Keep a backup of important files in a hard drive.

Risk Assessment

Lab Induction and Health and Safety Training completed.
Relevant COSHH and BioCOSHH Forms for lab work submitted.

Data Management Plan

Document completed and provided.

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